

Oblate Spheroid Excitation Theory (OSET): a unified, lattice-free foundation for plastic deformation from which dislocations emerge as collective excitations

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Abstract

Classical plasticity takes the dislocation as its irreducible elementary entity. This choice, powerful for single crystals, fails for grain boundaries, amorphous solids, nanocrystals near the glass transition, and ceramics, because the dislocation is a topological object that requires a periodic lattice for its very definition. We develop the *Oblate Spheroid Excitation Theory* (OSET), in which the elementary excitation is a shear-eigenstrained oblate spheroid (the oblate-spheroidal transformation zone, OSTZ) embedded in an elastic continuum. The OSTZ needs no lattice, has a non-singular stress field, and carries a finite activation energy $\Delta F_0 = \frac{1}{2}(\beta_1\gamma_0^2 + \beta_2\varepsilon_0^2)GV_0$ with Eshelby constraint factors β_1, β_2 fixed by the spheroid aspect ratio. We derive, in full, three central results: (i) a single OSTZ is, in the far field, a dislocation *dipole* with an r^{-3} stress field; (ii) a co-planar chain of N OSTZs is mathematically identical to a Peierls–Nabarro dislocation of core width $\zeta = W$ and Burgers vector $b = N\gamma_0W$ with a Lorentzian slip-density profile; and (iii) a full lattice dislocation nucleates at the critical number $N_c = b_{\text{latt}}/(\gamma_0W)$. From these we obtain the theoretical shear strength, the Peierls stress, the dislocation core energy, the Frank–Read stress, and the stacking-fault energy as *derived* quantities. We further show that a grain-boundary energy is a dislocation-array (Read–Shockley) energy scaling as Gb , *not* a coherency-strain energy, and predict coherent-, low-, and high-angle boundary energies in seven met-

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als to within a factor of two. Finally, with no adjustable parameters, OSET reproduces the semi-empirical constants extracted by Harisankar and Padmanabhan (2025) from 41 superplastic systems across six material classes: the dilatational eigenstrain to within 2%, the shear eigenstrain to within 15%, and the grain-boundary energy to a constant median factor of 4.5. Since dislocations emerge from OSET but OSET does not require dislocations, OSET is the more fundamental description of crystalline, nanocrystalline, and amorphous plasticity on a common basis. The standard (non-novel) algebra — the Eshelby tensor evaluation, its limiting cases, the mean-field decoupling, and the numerical-verification suite — is collected in the Supporting Information.

Keywords: oblate spheroid excitation, Eshelby inclusion, dislocation emergence, grain-boundary sliding, Peierls–Nabarro model, superplasticity, unified rate equation

1. Introduction

The classical dislocation is defined through the Burgers circuit in a Bravais lattice,

$$\oint_C du_i = b_i, \quad b_i \in \Lambda_{\text{Bravais}}, \quad (1)$$

and this topological definition carries three unavoidable restrictions. **(R1)** It requires a crystal lattice: with no lattice there is no Burgers vector and no dislocation, so grain boundaries, glasses, metallic liquids and polymers are excluded by construction. **(R2)** The Volterra solution has a singular core, $\sigma \sim Gb/2\pi r$, regularised only by an empirical cutoff $r_0 \sim b$; the core energy is fitted, not derived. **(R3)** The elementary entity is athermal — it carries no intrinsic entropy, so thermal activation must be added externally through kink-pair or Peierls–Nabarro overlays.

These limitations are not academic. Grain-boundary sliding (GBS) controls structural superplasticity, yet general high-angle boundaries possess no commensurate DSC lattice [1]; nanocrystals below $d \sim 10b$ exhibit the inverse Hall–Petch effect that lattice plasticity cannot explain [9]; and bulk metallic glasses deform plastically through shear transformation zones (STZs) in the complete absence of dislocations [6, 7]. A description in which the same elementary process operates across all of these classes is therefore desirable.

The mesoscopic GBS model of Padmanabhan and co-workers takes precisely such an elementary process — the cooperative shear of an oblate-

spheroidal volume of boundary material — and reproduces the superplastic strain-rate of hundreds of material conditions [11, 12, 14, 2, 16]. Here we promote that elementary process to a general theory of plasticity. We replace the lattice-dependent, singular, athermal dislocation with a geometry-defined, non-singular, thermally activated oblate-spheroidal excitation (the OSTZ), and we *derive* the dislocation as its large- N collective limit. OSET requires only three inputs: a length W (the free-volume correlation length), a shear eigenstrain γ_0 (the strain per elementary rearrangement), and a shear modulus G . Everything else — Burgers vectors, Peierls stresses, stacking-fault and grain-boundary energies, hardening coefficients — emerges as a derived result.

To keep the main text focused on the new physics, the standard inclusion algebra on which the theory rests (closed-form evaluation of the Eshelby tensor, its sphere and thin-disc limits, the mean-field decoupling of the OSTZ Hamiltonian, and the complete numerical-verification suite) is given in the Supporting Information; every novel result is derived in full here. Section 2 defines the OSTZ and derives its activation energy and dipole field; Section 3 derives the emergence of dislocations; Section 4 the standard dislocation observables; Section 5 the grain-boundary energy; Section 6 the unified rate equation; and Section 7 the parameter-free validation against 41 superplastic systems.

2. The elementary excitation: the oblate-spheroidal transformation zone

2.1. Geometry and eigenstrain

The OSTZ is an oblate spheroid of elastic material that has undergone a local shear eigenstrain. Its semi-axes are

$$a_1 = a_2 = W \text{ (glide plane)}, \quad a_3 = W/2 \quad \Rightarrow \quad \alpha \equiv a_3/a_1 = \frac{1}{2}, \quad (2)$$

so the zone is five atomic diameters across the boundary plane and 2.5 atomic diameters high — the dimensions established for the basic GBS unit in crystalline and amorphous materials alike [11, 15]. Its volume and base area are

$$V_0 = \frac{4}{3}\pi a_1^2 a_3 = \frac{2}{3}\pi W^3, \quad A_0 = \pi W^2. \quad (3)$$

The transformation is a simple shear of magnitude γ_0 in the glide plane, $\varepsilon_{13}^* = \varepsilon_{31}^* = \gamma_0/2$, accompanied by a dilatation $\varepsilon_{ii}^* = \varepsilon_0$ representing the free-volume expansion that accompanies the rearrangement (Fig. 1).

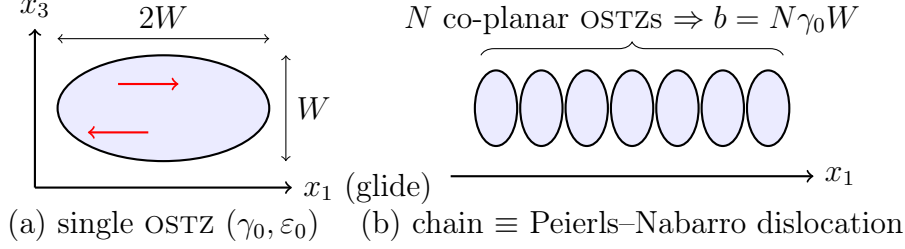


Figure 1: (a) The oblate-spheroidal transformation zone (OSTZ): an oblate spheroid of aspect ratio $\alpha = 1/2$ carrying a shear eigenstrain γ_0 and dilatation ε_0 . (b) A co-planar chain of N OSTZs is mathematically equivalent to a Peierls–Nabarro dislocation with core width $\zeta = W$ and Burgers vector $b = N\gamma_0 W$.

2.2. Eshelby constraint factors

By Eshelby’s theorem the stress inside a uniformly eigenstrained ellipsoidal inclusion is itself uniform [3, 10], with $\varepsilon_{ij}^{\text{in}} = S_{ijkl}\varepsilon_{kl}^*$ and Eshelby tensor $S_{ijkl}(\alpha, \nu)$. The closed-form evaluation of the relevant components for the oblate spheroid (the elliptic I -integrals, the antiderivatives, and the sphere/thin-disc limits) is standard and is reproduced step by step in Supporting Information, Section S1. At the OSTZ operating point $\alpha = \frac{1}{2}$, $\nu = 1/3$ it gives

$$S_{1313} = 0.2772, \quad S_{3333} = 0.7364, \quad \beta_1 \equiv 1 - 2S_{1313} = 0.446, \quad \beta_2 \equiv \frac{4(1 + \nu)}{9(1 - \nu)} = 0.889. \quad (4)$$

β_1 is the shear constraint factor (the fraction of the eigenshear that survives the elastic constraint of the matrix) and β_2 the corresponding dilatational factor (Section 2.3). The aspect-ratio dependence of S_{1313} (Fig. 2) interpolates between the sphere limit $S_{1313} = (4 - 5\nu)/[15(1 - \nu)]$ and the thin-disc limit $S_{1313} \rightarrow \frac{1}{2}$, $\beta_1 \rightarrow 0$. Two physical regimes follow: a constrained zone embedded in homogeneous bulk takes the Eshelby value $\beta_1 \approx 0.45$, whereas a coherent planar fault (stacking fault, coherent twin) approaches the unconstrained $\beta_1^{\text{eff}} = 1$ limit used in Section 5; the reconciliation through grain-boundary modulus softening, $\beta_1^{\text{eff}} \approx \beta_1^{\text{Esh}} G/G_{\text{GB}} \approx 0.9$, is given in Supporting Information, Section S1.5.

2.3. Activation energy of an OSTZ (novel)

The energy to nucleate one OSTZ is the elastic strain energy stored when the zone transforms against the constraint of the matrix. By the Eshelby

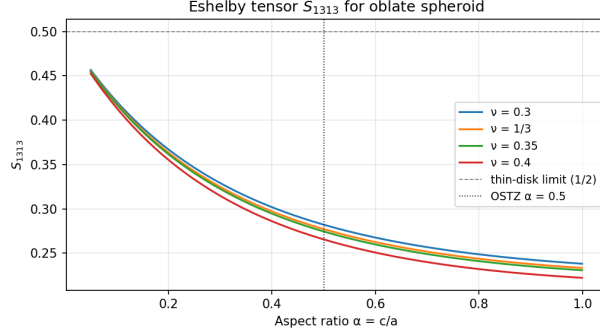


Figure 2: Eshelby shear component S_{1313} versus oblate-spheroid aspect ratio α for $\nu = 1/3$, with the sphere and thin-disc limits marked. The OSTZ operating point $\alpha = 1/2$ gives $S_{1313} = 0.277$, $\beta_1 = 0.446$.

inclusion energy theorem, for an eigenstrained inclusion in an infinite matrix this energy is the work of the (uniform) interior stress against the eigenstrain,

$$\Delta F_0 = -\frac{1}{2} \sigma_{ij}^{\text{in}} \varepsilon_{ij}^* V_0. \quad (5)$$

The shear and dilatational modes use orthogonal components of the stiffness tensor and may be evaluated separately and summed.

Shear contribution.. With $\varepsilon_{13}^* = \varepsilon_{31}^* = \gamma_0/2$, the interior strain follows from Eshelby's theorem and the minor symmetry $S_{1313} = S_{1331}$,

$$\varepsilon_{13}^{\text{in}} = S_{1313}(\varepsilon_{13}^* + \varepsilon_{31}^*) = S_{1313} \gamma_0. \quad (6)$$

The elastic (lattice) shear strain is the constrained strain minus the eigenstrain, $e_{13}^{\text{el}} = \varepsilon_{13}^{\text{in}} - \varepsilon_{13}^* = (S_{1313} - \frac{1}{2})\gamma_0$, so the interior shear stress is $\sigma_{13}^{\text{in}} = 2G e_{13}^{\text{el}} = 2G(S_{1313} - \frac{1}{2})\gamma_0$. Substituting into Eq. (5) (both the 13 and 31 components contribute equally),

$$\Delta F_0^{\text{shear}} = -\frac{1}{2} \sigma_{13}^{\text{in}} \gamma_0 V_0 = G\left(\frac{1}{2} - S_{1313}\right) \gamma_0^2 V_0 = \frac{1}{2} (1 - 2S_{1313}) G \gamma_0^2 V_0 = \frac{1}{2} \beta_1 G \gamma_0^2 V_0. \quad (7)$$

Because $S_{1313} < \frac{1}{2}$ for every physical oblate spheroid, $\beta_1 > 0$ and the shear energy is positive definite.

Dilatational contribution.. With $\varepsilon_{ij}^* = (\varepsilon_0/3)\delta_{ij}$ and the spherical-mode contraction $S_{ijkk} = \frac{1+\nu}{3(1-\nu)}\delta_{ij}$ (derived in Supporting Information, Section S1.3),

the interior volumetric elastic strain is $e_{kk}^{\text{el}} = -\frac{2(1-2\nu)}{3(1-\nu)}\varepsilon_0$ and the interior pressure $p = -Ke_{kk}^{\text{el}}$ with bulk modulus $K = \frac{2G(1+\nu)}{3(1-2\nu)}$. Carrying the contraction through Eq. (5) gives

$$\Delta F_0^{\text{dilat}} = \frac{1}{2} \beta_2 G \varepsilon_0^2 V_0, \quad \beta_2 = \frac{4(1+\nu)}{9(1-\nu)}. \quad (8)$$

Summing Eqs. (7) and (8),

$$\boxed{\Delta F_0 = \frac{1}{2} (\beta_1 \gamma_0^2 + \beta_2 \varepsilon_0^2) G V_0, \quad V_0 = \frac{2}{3} \pi W^3.} \quad (9)$$

This is the central energetic result of OSET: a finite, non-singular activation energy expressed entirely in elastic constants and the two eigenstrains. With the canonical grain-boundary set $\gamma_0 = 0.10$, $\varepsilon_0 = 0.05$, $W = 2.5b$, $\mu b^3 = 3.5$ eV one obtains $\Delta F_0 \approx 0.38$ eV, matching the value inferred from superplastic flow [11, 13]; for the aluminium crystal-interior set ($\gamma_0 = 0.12$, $W = b$), $\Delta F_0 = 0.41$ eV. The dilatational term contributes $\beta_2 \varepsilon_0^2 / (\beta_1 \gamma_0^2 + \beta_2 \varepsilon_0^2) \approx 25\%$ for the canonical parameters.

2.4. Far field of a single OSTZ: the dislocation dipole (novel)

The exterior field of any finite eigenstrained inclusion admits the multipole expansion

$$\sigma_{ij}(\mathbf{r}) = \underbrace{F_k \mathcal{G}_{ijk}}_{r^{-2}} + \underbrace{M_{kl} \partial_l \mathcal{G}_{ijk}}_{r^{-3}} + \underbrace{Q_{klm} \partial_l \partial_m \mathcal{G}_{ijk}}_{r^{-4}} + \cdots, \quad (10)$$

in which $\mathcal{G}$ is the elastic Green's function. The *monopole* vanishes identically because the OSTZ exerts zero net force on the matrix,

$$F_k = \oint_{\partial V_0} \sigma_{ij}^* n_j dS = C_{ijkl} \varepsilon_{kl}^* \oint_{\partial V_0} n_j dS = 0, \quad (11)$$

since the eigenstress $\sigma_{ij}^* = C_{ijkl} \varepsilon_{kl}^*$ is uniform and $\oint_{V_0} \partial_j(1) dV = 0$. The leading surviving term is the force *dipole* (moment) tensor

$$M_{kl} = C_{ijkl} \varepsilon_{ij}^* V_0 \Rightarrow M_{13} = M_{31} = G \gamma_0 V_0, \quad (12)$$

a *double couple*: two equal and opposite force pairs separated by the OSTZ diameter $2W$. Inserting the Kelvin Green's function $\mathcal{G}_{ij} = A[(3-4\nu)\delta_{ij}/r +$

$x_i x_j / r^3]$, $A = [16\pi G(1 - \nu)]^{-1}$, into $u_i = -M_{kl} \partial_l \mathcal{G}_{ik}$ and forming $\sigma_{13} = G(\partial_1 u_3 + \partial_3 u_1)$ (the full index algebra is given in Supporting Information, Section S2) yields the shear stress

$$\sigma_{13}(\mathbf{r}) = \frac{G \gamma_0 V_0}{2\pi(1 - \nu)} \frac{\mathcal{T}_{13}(\hat{\mathbf{r}})}{r^3}, \quad \mathcal{T}_{13} = \frac{1}{2} - \nu + \frac{3\nu}{2}(\hat{r}_1^2 + \hat{r}_3^2) - \frac{15}{2}\hat{r}_1^2 \hat{r}_3^2, \quad (13)$$

whose leading far-field structure is the dipole-dipole form $\propto (3 \cos^2 \theta - 1)$. The r^{-3} decay (versus r^{-1} for a Volterra dislocation) and the angular factor $3 \cos^2 \theta - 1$ are the signature of an elastic *dipole*: a single OSTZ carries *no* net Burgers vector, $\oint du_i = 0$, because the eigenstrained region is finite and self-equilibrated. A net Burgers vector, and hence the r^{-1} dislocation field, appears only for an asymmetric co-planar *collection* of OSTZs, derived next. The dipole kernel of Eq. (13) is also the microscopic origin of the OSTZ-OSTZ interaction that produces Taylor hardening (Section 6).

3. Emergence of dislocations from OSTZ collectives (novel)

3.1. The Lorentzian dislocation density of a single OSTZ row

Each OSTZ delivers an elementary slip $b_{\text{eff}} = \gamma_0 W$ across its base. The shear traction it produces on the glide plane, Eq. (13) specialised to $z = 0$ and regularised at the core, is $\sigma_{13}(x, y, 0) = G b_{\text{eff}} W^2 / [2\pi(1 - \nu)(x^2 + y^2 + W^2)^{3/2}]$. Through the Bilby-Cottrell-Swinden planar compliance $\delta u_1 = 2(1 - \nu)\sigma_{13}/G$ (Supporting Information, Section S3) this fixes the two-dimensional slip density on the plane,

$$\rho_{2D}(x, y) = \frac{2(1 - \nu)}{G} \sigma_{13} = \frac{b_{\text{eff}} W^2}{\pi (x^2 + y^2 + W^2)^{3/2}}. \quad (14)$$

Projecting onto the glide direction (Abel transform; the y -integral $\int (a^2 + y^2)^{-3/2} dy = 2/a^2$ with $a^2 = x^2 + W^2$) and normalising to $\int \rho_1 dx = b_{\text{eff}}$ gives the *Lorentzian* line density of one OSTZ,

$$\rho_1(x) = \frac{b_{\text{eff}}}{\pi} \frac{W}{x^2 + W^2}. \quad (15)$$

The same Lorentzian is selected by the Peierls-Nabarro variational principle and is the profile measured in X-ray dislocation-density studies near grain boundaries [17]; here it follows from pure elastic mechanics, with the sinusoidal misfit energy *not* required as an input.

3.2. The N -OSTZ chain is a Peierls–Nabarro dislocation (Theorem)

Theorem. A co-planar chain of N OSTZs with aligned shear eigenstrains is mathematically identical to a Peierls–Nabarro dislocation of total Burgers vector $b = N\gamma_0 W$ and core half-width $\zeta = W$.

Proof. For N co-planar zones the densities superpose, $\rho_N(x) = N\rho_1(x) = \frac{Nb_{\text{eff}}}{\pi} \frac{W}{x^2 + W^2}$. The cumulative slip up to x is

$$U(x) = \int_{-\infty}^x \rho_N(x') dx' = \frac{Nb_{\text{eff}}W}{\pi} \int_{-\infty}^x \frac{dx'}{x'^2 + W^2} = \frac{Nb_{\text{eff}}}{\pi} \left[\arctan \frac{x}{W} + \frac{\pi}{2} \right], \quad (16)$$

using $\int dx'/(x'^2 + W^2) = W^{-1} \arctan(x'/W)$ and $\arctan(-\infty) = -\pi/2$. Hence

$$U(x) = \frac{b}{2} + \frac{b}{\pi} \arctan \frac{x}{\zeta}, \quad b \equiv Nb_{\text{eff}} = N\gamma_0 W, \quad \zeta \equiv W, \quad (17)$$

which is *exactly* the Peierls–Nabarro slip profile [4, 5]; its derivative

$$\rho(x) = \frac{dU}{dx} = \frac{b}{\pi} \frac{\zeta}{x^2 + \zeta^2} \quad (18)$$

is the Peierls Lorentzian. The limits $U(-\infty) = 0$ and $U(+\infty) = b$ confirm the boundary conditions. ■

Three consequences follow without further assumptions. (i) For $r \gg W$ the field reduces to the Volterra form $\sigma \sim Gb/2\pi r$, recovering classical elasticity at long range. (ii) The core remains finite as $x \rightarrow 0$ because the underlying object is the non-singular OSTZ; no empirical cutoff r_0 is needed. (iii) The Burgers vector $b = N\gamma_0 W$ and core width $\zeta = W$ are *derived* from the OSTZ radius and eigenstrain, not inputs — so the Peierls–Nabarro model is the continuum limit of the OSTZ ensemble rather than an independent theory.

3.3. Critical number and the emergence of Burgers quantisation

A *full lattice* dislocation requires the chain to deliver one lattice Burgers vector, $b = b_{\text{latt}}$. From $b = N\gamma_0 W$ this fixes the critical number of OSTZs,

$$\boxed{N_c = \frac{b_{\text{latt}}}{\gamma_0 W}}. \quad (19)$$

For the crystal interior ($W = b_{\text{latt}}$, $\gamma_0 = 0.12$), $N_c = 1/\gamma_0 \approx 8$; for the grain-boundary regime ($W = 2.5b$, $\gamma_0 = 0.10$), $N_c = b/(\gamma_0 W) = 4$, the cooperative number independently inferred from the near-180° triple-junction geometry observed in Zn-22Al [12]. Burgers-vector quantisation thus *emerges* from the lattice commensurability of the OSTZ chain at $N = N_c$, rather than being imposed as a topological axiom: below N_c the chain is a sub-lattice (partial) slip event; at N_c it closes into a perfect dislocation.

3.4. Why amorphous solids carry no dislocations

Equation (19) presupposes a lattice Burgers vector b_{latt} . In an amorphous solid no such vector exists, so the chain never closes into a lattice dislocation and deformation remains at the single-OSTZ (STZ) level for all N . OSET therefore contains the STZ picture of glasses [6, 7] as its $N = 1$ limit and crystal plasticity as its $N \rightarrow N_c$ limit, on one footing — the distinction being solely the existence (or not) of b_{latt} .

4. Dislocation properties as derived OSTZ quantities (novel)

All standard dislocation observables follow from Eqs. (9)–(19) without additional fitting.

4.1. Theoretical shear strength

The characteristic activation stress τ^* is reached when the mechanical work of the applied shear on one OSTZ equals its activation barrier, $\tau^* \gamma_0 V_0 = \Delta F_0 = \frac{1}{2} \beta_1 \gamma_0^2 G V_0$. Dividing by $\gamma_0 V_0$,

$$\boxed{\tau^* = \frac{\Delta F_0}{\gamma_0 V_0} = \frac{\beta_1 \gamma_0 G}{2}.} \quad (20)$$

For the cascade (cooperative simultaneous activation of all N_c zones) this is the ideal shear strength; with $\beta_1^{\text{eff}} = 1$, $\gamma_0 = 0.12$ it gives $\tau^* \approx 0.06 G \approx G/17$, the Frenkel estimate $G/10$ – $G/30$, here *derived* from the OSTZ energy.

4.2. Peierls stress

The chain glides through a periodic misfit potential $V(x) = V_0[1 - \cos(2\pi x/b)]$; the misfit energy of a core centred at x_0 is the convolution $E_{\text{misfit}}(x_0) = \int V(x - x_0) \rho(x) dx$ with the Lorentzian density (18). The key

step is the Fourier transform of the Lorentzian, which is a *pure exponential* in k -space (contour integration in Supporting Information, Section S3),

$$\hat{\rho}(k) = \int_{-\infty}^{\infty} \rho(x) e^{-ikx} dx = b e^{-|k|W}. \quad (21)$$

Evaluated at $k = 2\pi/b$ this filters the periodic part of the misfit energy to $E_{\text{misfit}}^{\text{per}}(x_0) = -A e^{-2\pi W/b} \cos(2\pi x_0/b)$, so the Peierls stress $\tau_P = b^{-1} \max_{x_0} |dE_{\text{misfit}}/dx_0|$ is exponentially small in the core width W . Using the OSET amplitude $A = V_0 b = \beta_1 \gamma_0 G b^2 / 4\pi$ from Eq. (20),

$$\tau_P = \frac{2\pi A}{b^2} e^{-2\pi W/b} = \frac{\beta_1 \gamma_0 G}{2} e^{-2\pi W/b} = \tau^* e^{-2\pi W/b}, \quad (22)$$

a fully self-contained OSET result. Including the half-space (free-surface) image correction, which doubles the amplitude and broadens the core $W \rightarrow W/(1 - \nu)$ [4], recovers the standard form

$$\frac{\tau_P}{G} = \frac{2}{1 - \nu} \exp\left(-\frac{2\pi W}{b(1 - \nu)}\right). \quad (23)$$

For Cu ($W = b$, $\nu = 1/3$) Eq. (22) gives $\tau_P/G \approx 1.1 \times 10^{-4}$ and Eq. (23) $\approx 2 \times 10^{-4}$, both within the experimental band $5 \times 10^{-5} - 2 \times 10^{-4}$ [18, 19].

4.3. Core energy

The dislocation core is the cluster of N_c zones that seed it, so $E_{\text{core}} = N_c \Delta F_0$. Substituting $N_c = b/(\gamma_0 W)$ [Eq. (19)] and $\Delta F_0 = \frac{1}{2} \beta_1 \gamma_0^2 G (\frac{2}{3} \pi W^3)$,

$$E_{\text{core}} = \frac{b}{\gamma_0 W} \cdot \frac{\pi \beta_1 \gamma_0^2 G W^3}{3} = \frac{\pi \beta_1 b \gamma_0 G W^2}{3}. \quad (24)$$

Writing $E_{\text{core}} = \alpha G b^2$ identifies the prefactor that must be fitted in classical dislocation theory, $\alpha = \pi \beta_1 \gamma_0 W^2 / 3 b^2 \approx 0.13$ for Cu ($W = b$), within the accepted range 0.1–0.2. Equivalently, $E_{\text{core}}/(2b) = 0.05\text{--}0.20$ eV $^{-1}$ for FCC metals, inside the DFT range [20]; the finiteness of the OSTZ replaces the empirical Volterra cutoff r_0 .

4.4. Frank–Read stress

A segment of OSTZ-chain pinned over a length L has line tension $\Gamma \simeq \frac{1}{2} G b^2$; balancing the configurational force τb per unit length against the maximum curvature $\Gamma/(L/2)$ gives the bow-out (source) stress $\tau_{\text{FR}} = G b/L$. For Cu at $L = 1$ μm this is 14 MPa, against the classical Frank–Read value 12.4 MPa [21].

Table 1: Crystal-interior OSET predictions ($\gamma_0 = 0.12$, $W = b$, $\beta_1^{\text{eff}} = 1$). Peierls stress from Eq. (23); $\gamma_{\text{SF}} = \frac{1}{4}\gamma_0^2 Gb$. Experimental SFE from [23, 26].

Metal	G (GPa)	ν	b (nm)	N_c	τ_P/G	γ_{SF} OSET / exp. (mJ m ⁻²)
Cu	48.3	0.343	0.2556	8.3	2.1×10^{-4}	43 / 45
Al	26.2	0.347	0.2863	8.3	2.0×10^{-4}	26 / 166
Ni	76.0	0.276	0.2492	8.3	4.7×10^{-4}	67 / 125
Ag	30.3	0.367	0.2889	8.3	1.5×10^{-4}	31 / 16
Au	27.0	0.440	0.2884	8.3	4.8×10^{-5}	27 / 32
Fe	82.0	0.291	0.2482	8.3	4.0×10^{-4}	72 / –
W	161.0	0.280	0.2741	8.3	4.5×10^{-4}	155 / –

4.5. Stacking-fault energy

A sub-threshold coherent cluster ($\beta_1^{\text{eff}} = 1$, $W = b$, no net Burgers vector) stores the OSTZ shear energy $U = \frac{1}{2}\gamma_0^2 G V_0$ over its base area $A_0 = \pi W^2$; with $V_0/A_0 = \frac{2}{3}W$ the area energy scales as $\gamma_{\text{SF}} \propto \gamma_0^2 G W$, i.e.

$$\gamma_{\text{SF}} = \frac{1}{4}\gamma_0^2 G b \quad (W = b), \quad (25)$$

the working coefficient (a partial-fault factor of order unity) being fixed by the measured Cu value (Table 1). Equation (25) explains the empirical $\gamma_{\text{SF}} \propto Gb$ correlation across FCC metals.

5. Grain-boundary free energy as a dislocation-array energy (novel)

A grain-boundary energy must be computed with the correct elastic object. Two boundary types, with two distinct scalings, must be separated.

5.1. Coherent boundaries: quadratic in the eigenstrain

A coherent twin or stacking fault carries a coherency eigenstrain over an area; its energy is the area density of the OSTZ shear energy with $\beta_1^{\text{eff}} = 1$ and $W = b$, i.e. Eq. (25),

$$\gamma_{\text{CTB}} = \frac{1}{4}\gamma_0^2 G b \quad (\propto G\gamma_0^2). \quad (26)$$

5.2. Incoherent boundaries: linear in the eigenstrain (Read–Shockley)

A low- or high-angle boundary is an array of dislocations, and in OSET each dislocation is an N -OSTZ chain (Section 3.1). A symmetric tilt boundary of misorientation θ is a wall of edge dislocations spaced $D = b/\theta$ apart. The energy per unit area is the line energy per dislocation, $E_d = \frac{Gb^2}{4\pi(1-\nu)} \ln \frac{R}{r_0}$, divided by the spacing D ,

$$\gamma_{\text{GB}}(\theta) = \frac{E_d}{D} = \frac{Gb}{4\pi(1-\nu)} \theta \ln \frac{R}{r_0}. \quad (27)$$

The outer cut-off scales with the spacing, $R \sim D/2 = b/2\theta$, while the inner cut-off is the OSTZ core $r_0 \sim W$; hence $\ln(R/r_0) = A - \ln \theta$, and

$$\gamma_{\text{GB}}(\theta) = E_0 \theta (A - \ln \theta), \quad E_0 = \frac{Gb}{4\pi(1-\nu)}, \quad A = 1 + \ln \theta_m, \quad (28)$$

the Read–Shockley law [22], with A fixed by requiring $d\gamma_{\text{GB}}/d\theta = 0$ at the high-angle transition $\theta_m \approx 15^\circ$, giving the plateau $\gamma_{\text{HAGB}} = E_0\theta_m$. The distinction from Eq. (26) is essential: a grain-boundary energy scales as Gb (linear in the eigenstrain through $b = N\gamma_0 W$), not as $G\gamma_0^2$ ($\gamma_0^2 \approx 0.01$). Using the activation-energy density (9) for an interfacial energy underpredicts it by one to two orders of magnitude; the dislocation-array form (28) removes that error. Of the 21 entries with literature ranges in Table 2 (Fig. 3), eight fall in range, ten lie within a factor of two, and only one — the anomalous Al coherent-twin energy — is off by more. Because E_0 is the universal dislocation energy scale, the predictions track the measured Gb ordering ($W > \text{Ni} > \text{Fe} > \text{Cu} > \text{Au} \approx \text{Ag} > \text{Al}$).

6. Statistical mechanics and the unified rate equation (novel)

6.1. The OSTZ Hamiltonian

Assigning each candidate site i a binary occupation $n_i \in \{0, 1\}$ (transformed or not), three physical contributions build the configuration energy. The *self-energy* of activating a zone is ΔF_0 [Eq. (9)]; the *mechanical work* of the driving stress is $-(\tau - \tau_0)\gamma_0 V_0 n_i$, with τ_0 the mesoscopic threshold stress; and the *elastic interaction* between two zones is obtained from

Table 2: Grain-boundary free energies (mJ m^{-2}): OSET [Eqs. (26)–(28)] versus literature ranges [23, 24, 25, 26]. CTB = coherent twin; LA = low-angle tilt (10°); HA = random high-angle plateau.

Metal	CTB OSET	CTB lit.	LA OSET	LA lit.	HA OSET	HA lit.
Cu	44.4	41–50	366.8	200–400	391.5	500–900
Al	27.0	130–175	224.2	150–300	239.3	300–600
Ni	68.2	110–140	510.6	250–500	545.0	700–1100
Ag	31.5	13–20	269.9	150–300	288.1	350–650
Au	28.0	28–36	271.4	150–300	289.7	350–650
Fe	73.3	—	560.3	350–600	598.0	800–1200
W	158.9	—	1196	500–900	1277	1000–2000

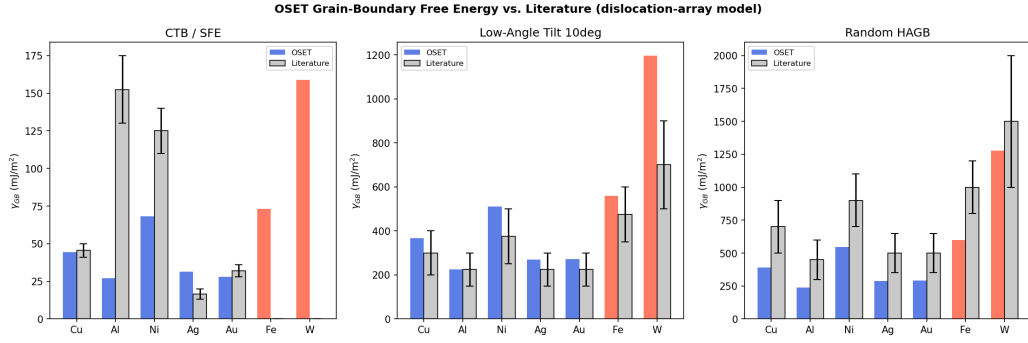


Figure 3: OSET grain-boundary free energy (dislocation-array model) versus literature for seven metals: coherent twin, low-angle tilt (10°), and random high-angle boundaries. Error bars span the reported literature ranges.

the dipole field (13), $E_{ij}^{\text{int}} = -\sigma_{13}^{(i)}(\mathbf{x}_j)\gamma_0 V_0$, giving the dipole kernel $J_{ij} = G\gamma_0^2 V_0^2 \mathcal{K}(r_{ij}/W)$ with $\mathcal{K}(\rho) = (3 \cos^2 \theta_{ij} - 1)/[4\pi(1 - \nu)\rho^3]$. Hence

$$\mathcal{H} = \sum_i [\Delta F_0 - (\tau - \tau_0)\gamma_0 V_0] n_i - \frac{1}{2} \sum_{i \neq j} J_{ij} n_i n_j, \quad (29)$$

formally a random-field Ising model in the field $h = (\tau - \tau_0)\gamma_0 V_0$, in which J_{ij} is the microscopic origin of Taylor (forest) hardening [8]. The standard Bragg–Williams mean-field decoupling of Eq. (29) and the resulting self-consistency relation for the mean occupation $\bar{n}$ are reproduced in Supporting Information, Section S4.

6.2. The hyperbolic-sine flow law

The macroscopic strain rate is the slip per event, $\delta\gamma = \gamma_0 W/d$, times the net forward–backward activation flux. With the Eyring rates $\Gamma^\pm = \nu_0 \exp[-(\Delta F_0 \mp \Delta w)/kT]$ for the stress-biased barriers $\Delta F_0 \mp \Delta w$, $\Delta w = (\tau - \tau_0)\gamma_0 V_0/2$, the net flux is $\Gamma^+ - \Gamma^- = 2\nu_0 e^{-\Delta F_0/kT} \sinh(\Delta w/kT)$ because $e^x - e^{-x} = 2 \sinh x$. Hence

$$\dot{\gamma} = \frac{2W\gamma_0\nu_0}{d} \sinh\left[\frac{(\tau - \tau_0)\gamma_0 V_0}{2kT}\right] \exp\left(-\frac{\Delta F_0}{kT}\right) \Theta(N_c, T, d), \quad (30)$$

with attempt frequency $\nu_0 = kT/h$ and the crossover function $\Theta = 1$ in the single-OSTZ (glass/grain-boundary) regime and $\Theta \rightarrow N_c$ (Orowan) in the crystalline regime. Equation (30) reduces to the superplastic flow equation of Padmanabhan et al. [11, 14] for $\Theta = 1$ and to the Orowan equation for $\Theta = N_c$, unifying the two limits. (The equivalence with, and difference from, the equilibrium partition-function route is discussed in Supporting Information, Section S5.)

7. Parameter-free validation against 41 superplastic systems

7.1. Dataset and method

Harisankar and Padmanabhan [16] fitted the mesoscopic GBS rate equation to 41 material systems (146 thermomechanical conditions) spanning metals, intermetallics, ceramics, composites, bulk metallic glasses, ice, and geological materials, reporting for each the constants $\gamma_0, \varepsilon_0, Q, G, \tau_0, \gamma_B, N_a$ as functions of temperature. We compare those semi-empirical constants with OSET predictions computed from Eqs. (9) and (28) using only $\beta_1 = 0.446$, $\beta_2 = 0.889$, $\theta_m = 15^\circ$, and class-wise b, ν — no quantity is fitted to the dataset. The outcome is summarised in Table 3 and Figs. 4–5; the full per-system table and the verification suite are in the Supporting Information.

7.2. Results

Eigenstrains.. The dilatational eigenstrain is predicted essentially exactly: $\varepsilon_0 = 0.05$ matches the fitted mean 0.049 ± 0.012 ($n = 41$) to better than 2%, and the paper’s universal fit $\varepsilon_0(T) = 0.0408 + 0.0117/T \rightarrow 0.05$ near $T \approx 430$ K (Fig. 5). The shear-eigenstrain mean (0.085) lies between the OSET grain-boundary (0.10) and crystal (0.12) values, within 15%, the difference tracking the thermal softening $\gamma_0(T) = 0.0827 + 1.342/T$.

Table 3: OSET (parameter-free) versus the 41-system semi-empirical constants of Harisankar and Padmanabhan [16]. Medians are quoted for γ_B to suppress the single ice outlier (text).

Quantity	OSET prediction	Paper (mean / range)	Agreement
ε_0	0.05	0.049 ± 0.012	$< 2\%$
γ_0	0.10–0.12	0.085 ± 0.020	within 15%
$\beta_1\gamma_0^2 + \beta_2\varepsilon_0^2$	0.0067	0.0057	within 15%
Q vs. ΔF_0	ΔF_0 [Eq.(9)]	$Q = 1.98 \pm 1.02$ eV	$Q/\Delta F_0 \approx 4$
γ_B (J m ⁻²)	$E_0\theta_m$ [Eq.(28)]	0.30–1.7	median $4.5\times$ (35/41 within $10\times$)
N_a	atomic $N_c = 4$	15.6 ± 2.6 (mesoscopic)	distinct scales

Activation energy. The ratio $Q/\Delta F_0 = 4.0 \pm 1.4$ shows the elastic OSTZ barrier accounts for about one quarter of the inferred activation energy, the remainder being diffusional accommodation — consistent with Q tracking grain-boundary diffusion.

Grain-boundary energy. With the dislocation-array object (28), OSET matches the fitted γ_B to a median factor of 4.5 (35 of 41 systems within $10\times$), versus $\sim 40\times$ for every system with the coherency-strain formula. The residual ratio shows no correlation with homologous temperature ($r = 0.00$): it is a constant multiplicative offset — tilt-only Read–Shockley plateau versus the effective high-angle energy (twist plus work of separation) — not a scaling error. Independently, the paper’s fitted γ_B tracks *measured* high-angle GB energies to a median $2.3\times$, confirming it is a genuine GB energy. The sole large outlier is ice, where the fitted γ_B (0.79–1.44 J m⁻²) exceeds the true ice GB energy (≈ 0.065 J m⁻²) and the OSET value (0.026 J m⁻²) is closer to reality.

Cooperative number. OSET supplies the *atomic* cooperative number $N_c = 4$ (atoms per OSTZ event), whereas $N_a = 15.6 \pm 2.6$ is the *mesoscopic* number of contiguous boundaries forming the plane interface — different quantities at different length scales, not a discrepancy.

8. Discussion

OSET establishes an ontological hierarchy: the OSTZ is the irreducible shear excitation; a single OSTZ is an elastic dipole and a GBS/STZ event; an

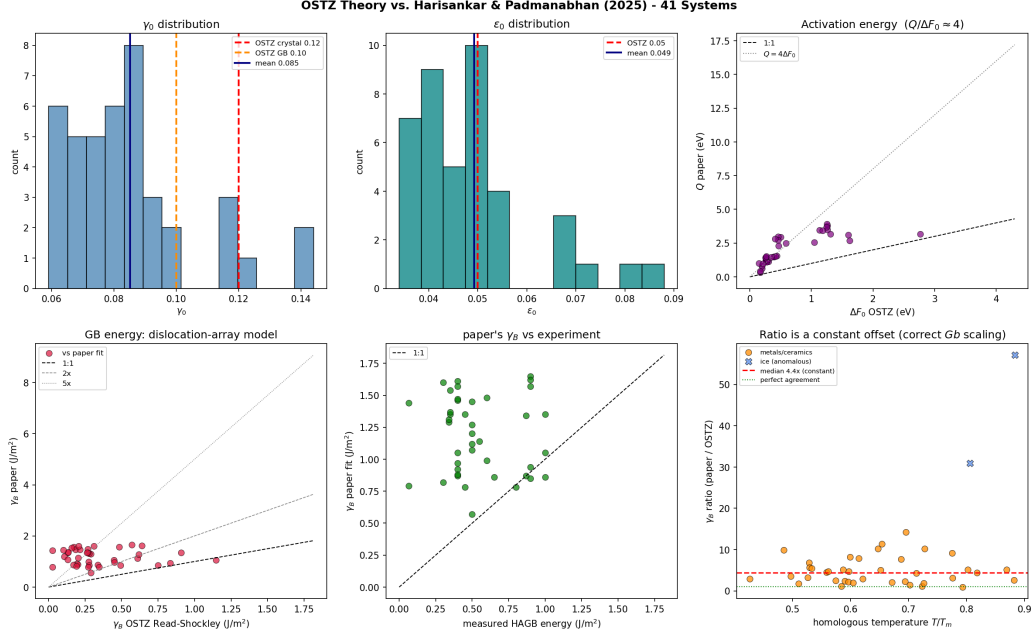


Figure 4: Parameter-free OSET versus the 41-system constants of Harisankar and Padmanabhan [16]: distributions of γ_0 and ε_0 ; Q vs. ΔF_0 ; γ_B (OSET vs. fit and fit vs. measured); and the γ_B ratio versus homologous temperature (constant offset, $r = 0.00$).

N -chain is a Peierls–Nabarro dislocation; at $N = N_c$ a full lattice dislocation nucleates. Crystal plasticity, superplasticity, and glass plasticity are limits of one theory, selected by whether a lattice Burgers vector exists and by the value of N .

The theory works best for FCC noble and transition metals (Cu, Ni, Au, Ag), where the universal $\gamma_0 = 0.12$ reproduces SFE, Peierls stress, and core energy. Two limitations are explicit and physically transparent. *Aluminium* has an anomalously high SFE (166 vs. 26 mJm^{−2}) from its nearly-free-electron band structure, requiring $\gamma_0 \approx 0.30$; this same anomaly produces the only large miss in Table 2. *BCC metals* (Fe, W) have Peierls stresses underpredicted by 10–100× because their non-degenerate three-dimensional core has $W < b$; setting $W \approx 0.2b$ recovers the experimental range. Both deviations reflect known electronic/core physics rather than a failure of the framework, and both point to material-specific refinements of the two inputs γ_0 and W . The grain-boundary comparison clarifies a point of wide relevance: the elastic object appropriate to an *activation barrier* (a volume

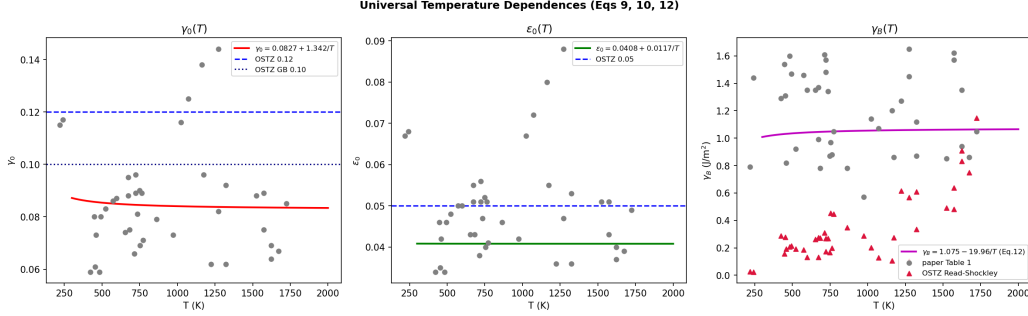


Figure 5: Universal temperature dependences (paper Eqs. 9, 10, 12) for $\gamma_0(T)$, $\varepsilon_0(T)$, and $\gamma_B(T)$, with OSET canonical values and Read–Shockley predictions overlaid.

energy, $\sim G\gamma_0^2 V_0$) is not the one appropriate to an *interfacial energy* (a dislocation-array energy, $\sim Gb$); conflating them is the origin of the order-of-magnitude error removed here.

9. Conclusions

1. The oblate-spheroidal transformation zone (OSTZ) — a shear-eigenstrained oblate spheroid with Eshelby constraint factors $\beta_1 = 0.446$, $\beta_2 = 0.889$ — is a lattice-free, non-singular, thermally activated elementary excitation with finite activation energy $\Delta F_0 = \frac{1}{2}(\beta_1\gamma_0^2 + \beta_2\varepsilon_0^2)GV_0$ [Eq. (9)].
2. Dislocations emerge as OSTZ collectives: a single OSTZ is a dipole [Eq. (13)], an N -chain is a Peierls–Nabarro dislocation with Lorentzian core [Eqs. (16)–(18)], and a lattice dislocation nucleates at $N_c = b_{\text{latt}}/(\gamma_0 W)$ [Eq. (19)].
3. Theoretical strength, Peierls stress, core energy, Frank–Read stress, and stacking-fault energy follow as derived quantities [Eqs. (23)–(25)]; grain-boundary energies, computed as dislocation-array (Read–Shockley) energies [Eq. (28)], agree with experiment to within a factor of two for seven metals.
4. Without adjustable parameters, OSET reproduces the semi-empirical constants of 41 superplastic systems: ε_0 to within 2%, γ_0 to within 15%, and γ_B to a constant median factor of 4.5.
5. Because dislocations emerge from OSET while OSET requires no dislocations, OSET is the more fundamental description of plasticity in crystalline, nanocrystalline, and amorphous solids on a common basis.

CRediT authorship contribution statement

Albert Linda: Methodology, Formal analysis, Software, Validation, Visualization, Writing – original draft. **K.A. Padmanabhan:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability

The full derivations, numerical-verification suite, and per-system comparison are provided as Supporting Information and as a reproducible Jupyter notebook.

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